

Amatya Katyayan

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Professional Summary

Robotics and AI graduate student at Arizona State University with hands-on experience building real-world intelligent systems across industry, research, and education. Published researcher and former software engineer specializing in computer vision, autonomous robotics, and deep learning. Driven to design reliable, scalable, and socially responsible systems that perform under real-world constraints through disciplined engineering and collaborative leadership.

Education

M.S. in Robotics and Autonomous Systems (Artificial Intelligence Track)	May 2027
Arizona State University, Tempe, AZ	GPA: 4.22 / 4.00
B.E. in Computer Engineering	May 2022
Goa University, Goa, India	86.48%

Technical Skills

Programming: Python, C++, Java, MATLAB, JavaScript
AI/ML: TensorFlow, PyTorch, Scikit-learn, Reinforcement Learning, MDPs
Computer Vision: OpenCV, YOLOv5, Object Detection, Image Segmentation
Robotics: ROS, Dobot SDK (pydobot, pydobotplus), Kinematics, Motion Planning
Web Dev: React, Spring Boot, REST APIs **Version Control:** Git, GitHub
Visualization: Matplotlib, OpenCV Visualization, Data Plotting

Professional Experience

Technical Trainer, 6 Technical Training Regiment (Indian Army)	<i>Aug 2024 - May 2025</i>
- Led technical training programs in AI, cybersecurity, and databases for Army personnel. - Designed hands-on learning modules integrating applied robotics and real-world scenarios. - Mentored trainees to improve operational readiness and technical confidence.	
Software Developer, Persistent Systems	<i>Aug 2022 - Jan 2024</i>
- Designed and deployed enterprise-grade React and Spring Boot platforms, reducing manual reporting workload by 40%. - Built REST APIs for AI-based skill-job matching and hiring analytics. - Collaborated with cross-functional engineering and product teams to deliver production-grade, data-driven systems under tight timelines.	

Selected Research & Robotics Projects

Autonomous Tic-Tac-Toe Playing Robot - (Demo Video)
<i>Computer Vision + AI + Dobot Magician Lite</i>
Built a system that detects human-drawn X/O symbols using OpenCV and contour-based vision modeling, responding with AI-driven moves through the Dobot arm.
Object Detection and Classification using Dobot & YOLO - (Demo Video)
Created an automated pick-and-place robot using YOLOv5 for real-time object classification and robotic sorting through motion planning.
AI-Driven Maze Solving Robot - (Demo Video)
Developed an AI system that perceives maze structures via a top-mounted camera and guides the Dobot arm to trace the solution path using perception and control integration.
Smart Agriculture System using AI and IoT - (Demo Video)
Designed an adaptive farm-assist tool that analyzes sensor data for personalized crop insights. Funded by Goa State Innovation Council and published in IEEE Xplore .

Publication

Design of Smart Agriculture Systems using Artificial Intelligence and Big Data Analytics - Amatya Katyayan, [IEEE Xplore](#), 2022

Achievements & Activities

Student Experience Assistant, ISSC, ASU: Supported international arrivals, airport reception, and onboarding.
[IEEE CSPL 2021](#) - *First Place* • Texas Instruments India Innovation Contest: Quarter Finalist • College Topper, Goa University • Campus Ambassador, TCS • Creator of "[Amatya Katyayan](#)" Tech YouTube Channel